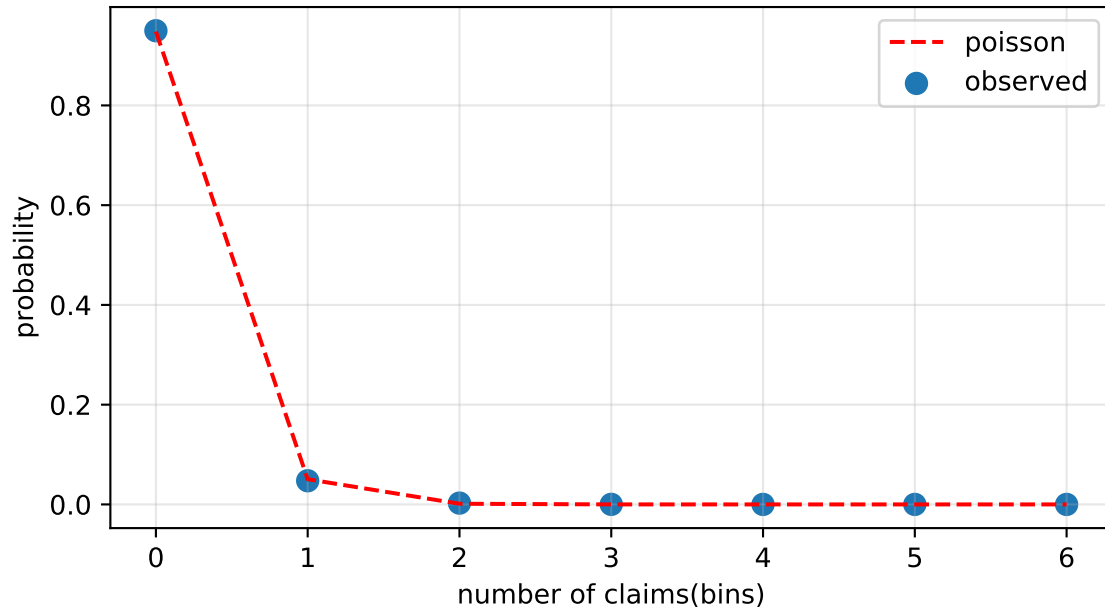
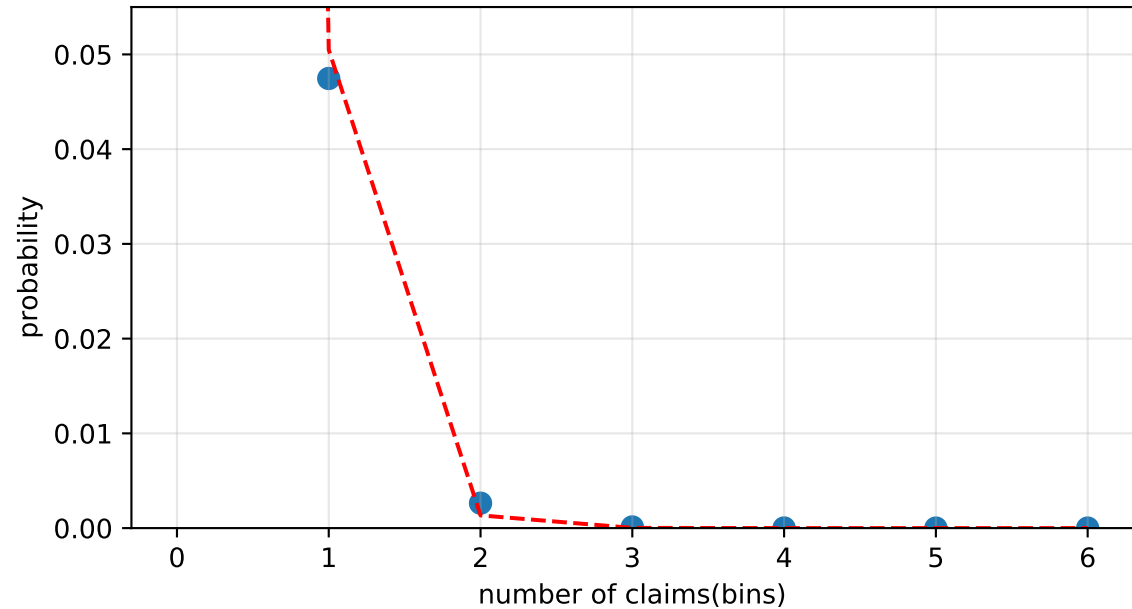


comparison: empirical data vs. poisson distribution

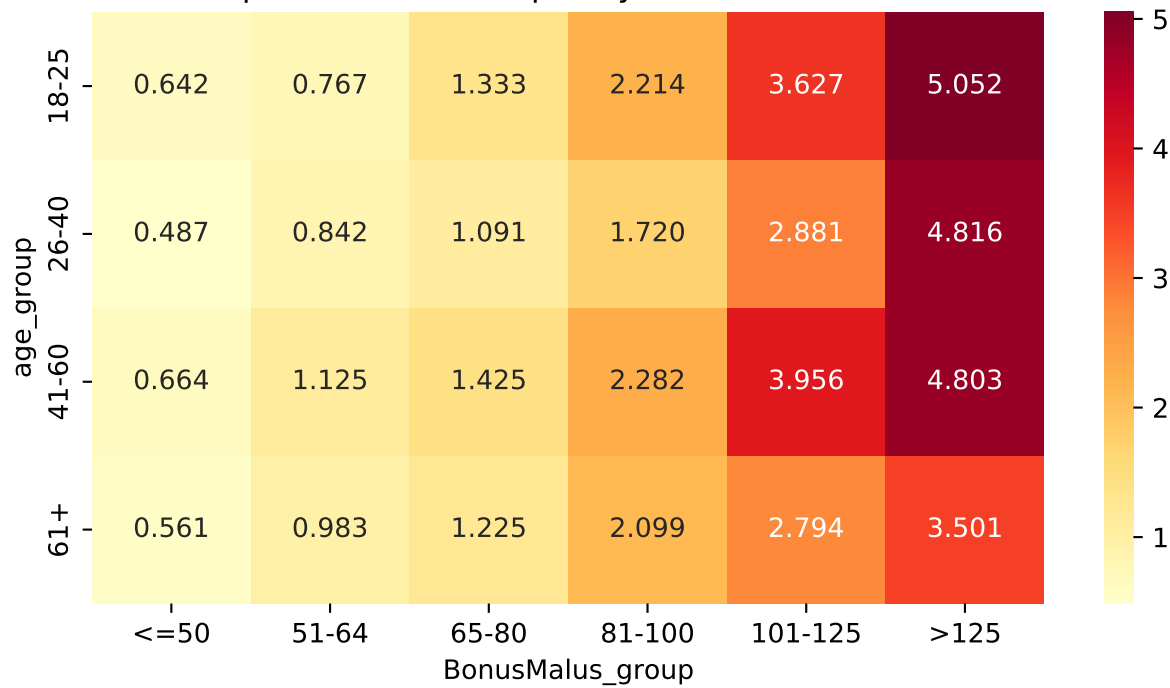
complete overview



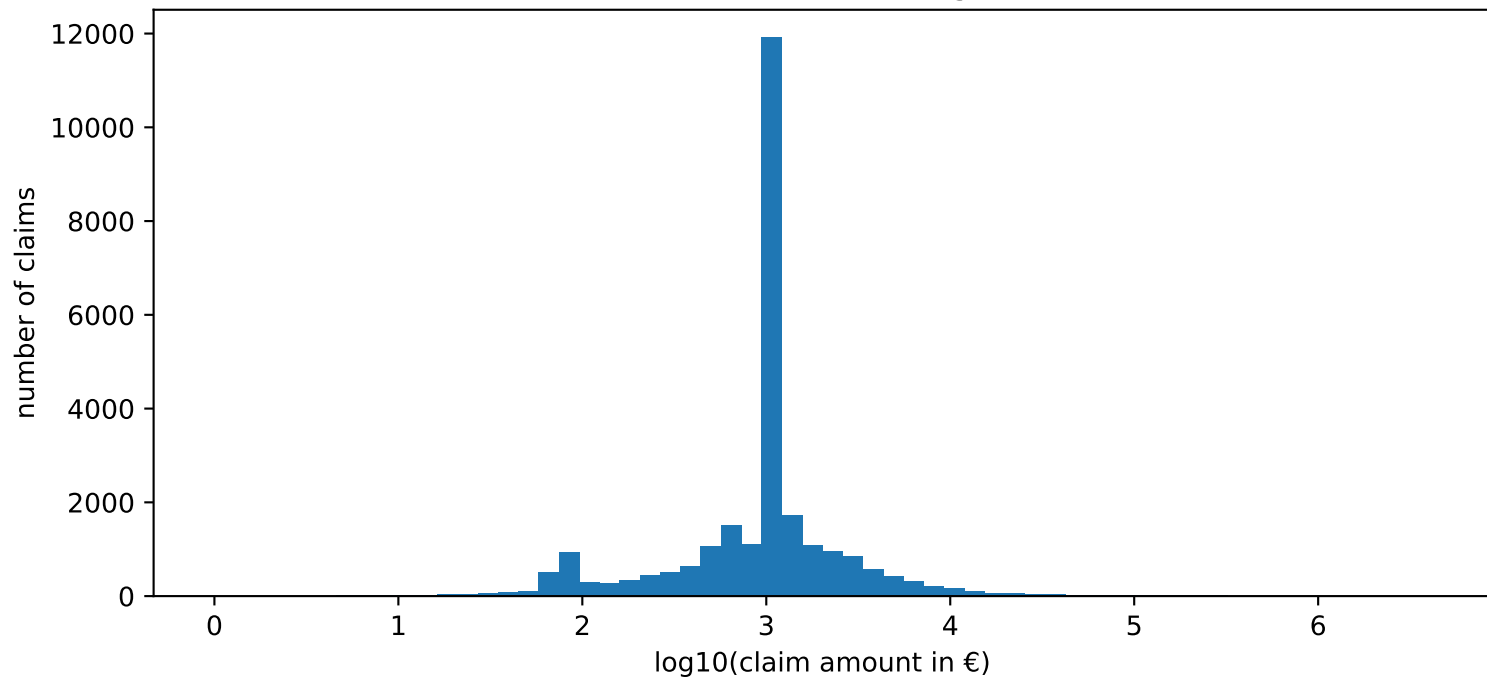
zoom on rare damage events



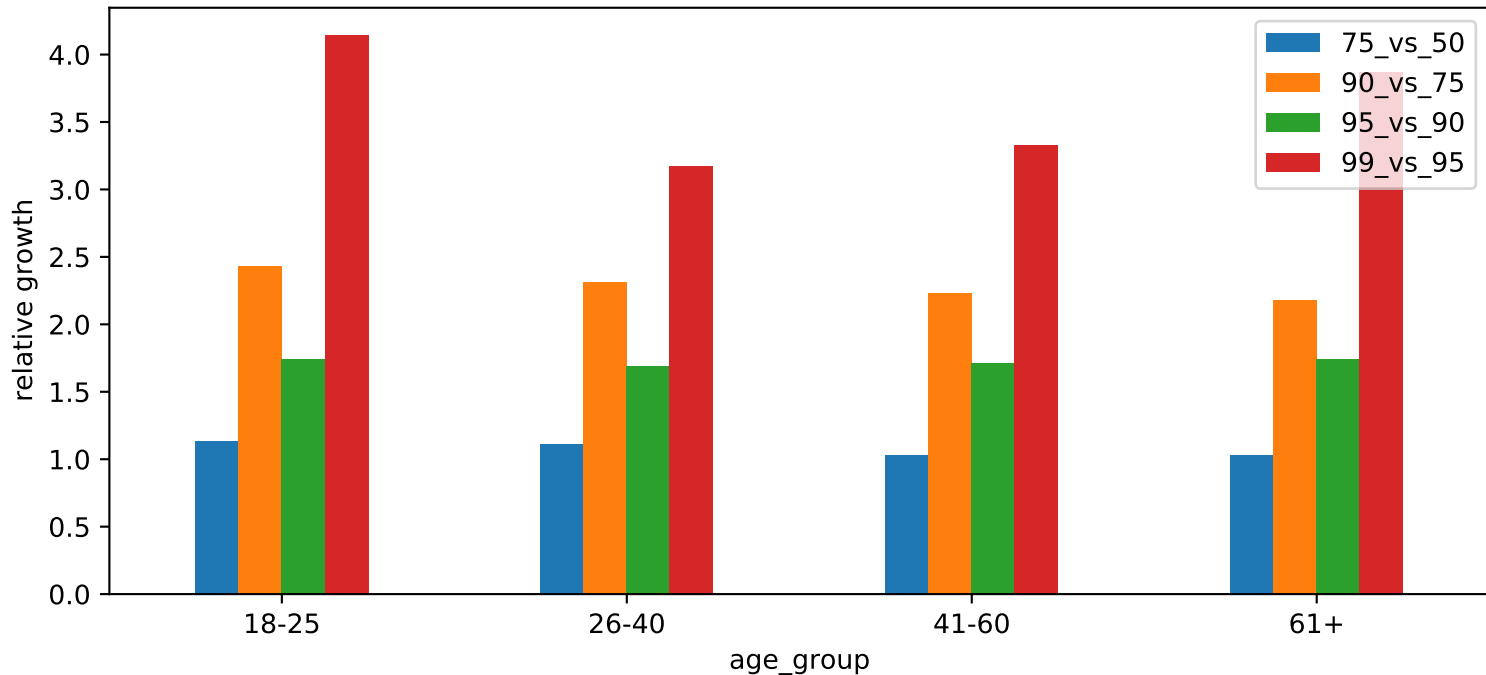
expected claims frequency from Poisson-GLM



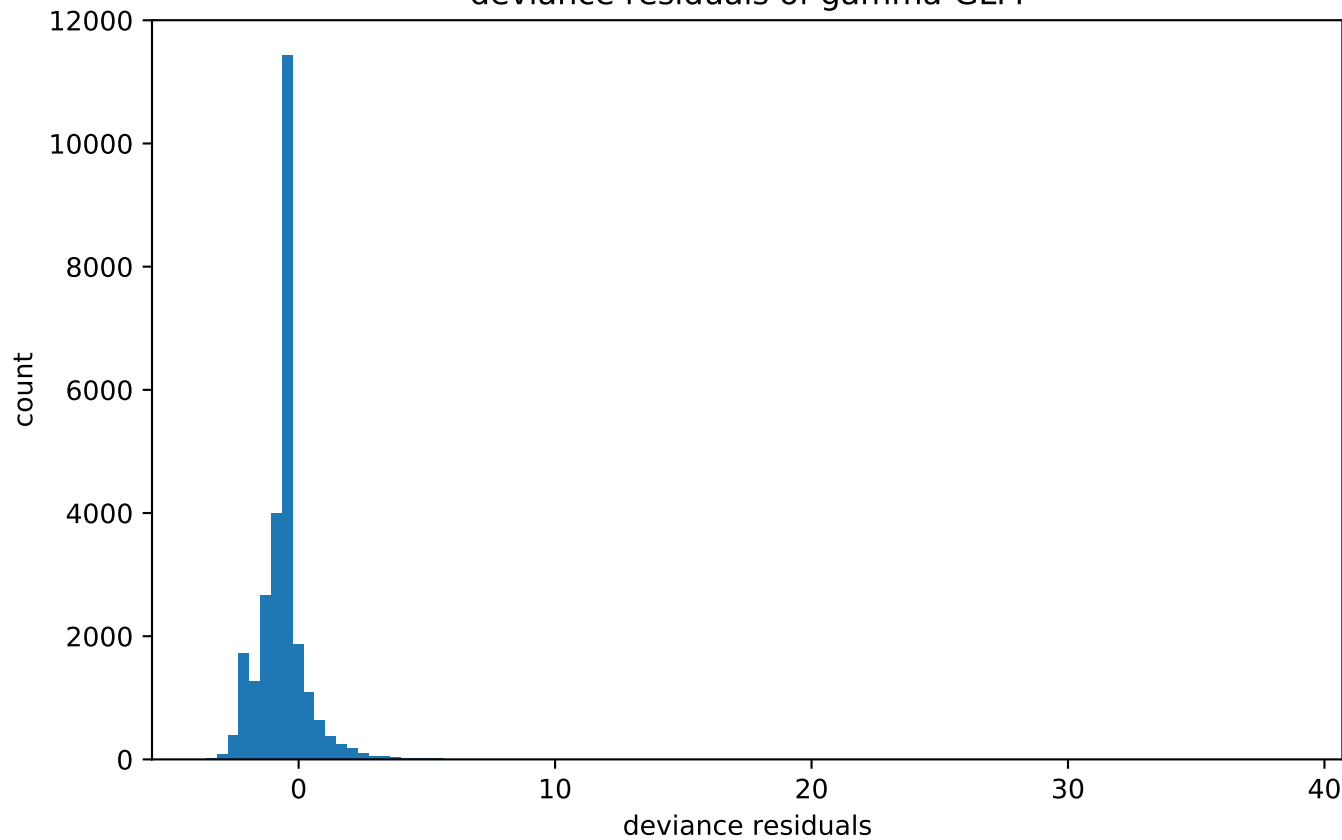
distribution claim amount on logarithmic scale



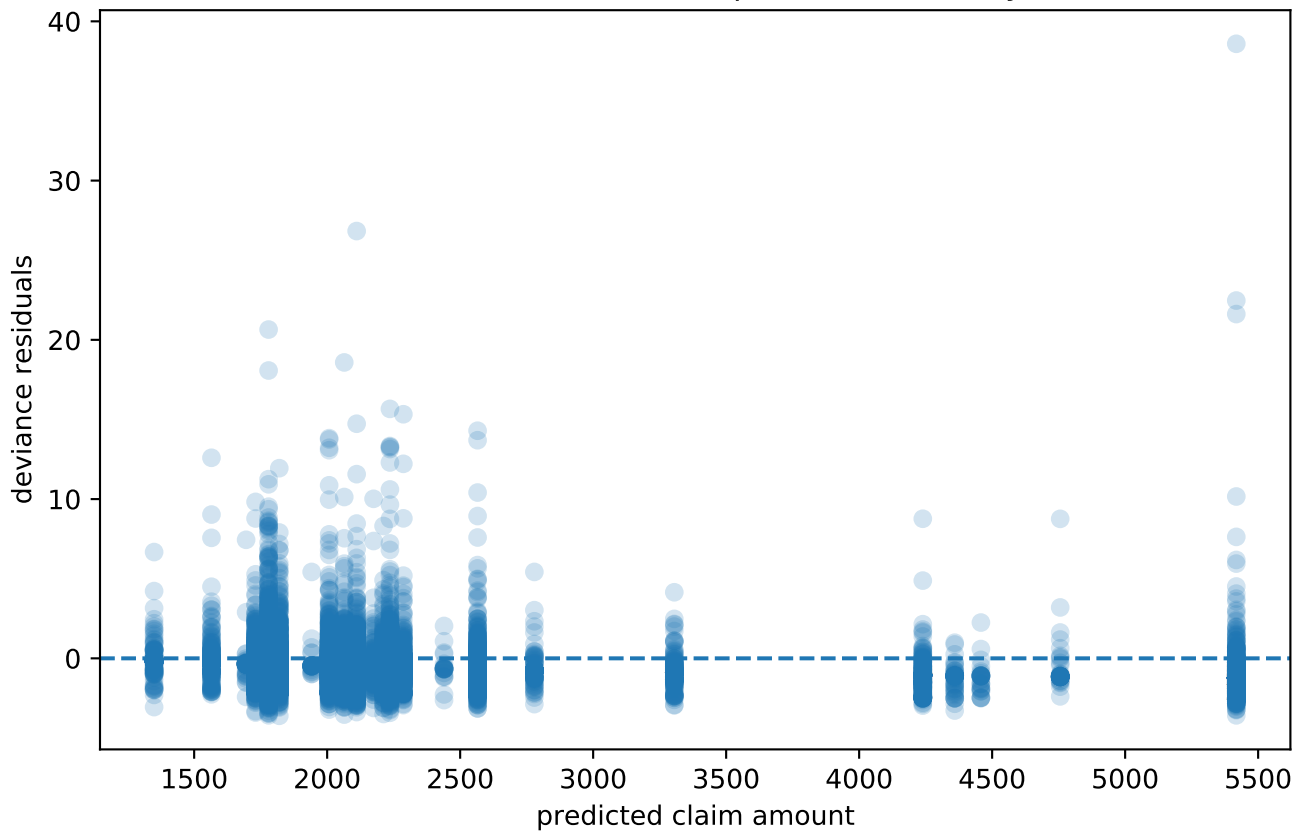
tail growth of claim amount sorted after age_group



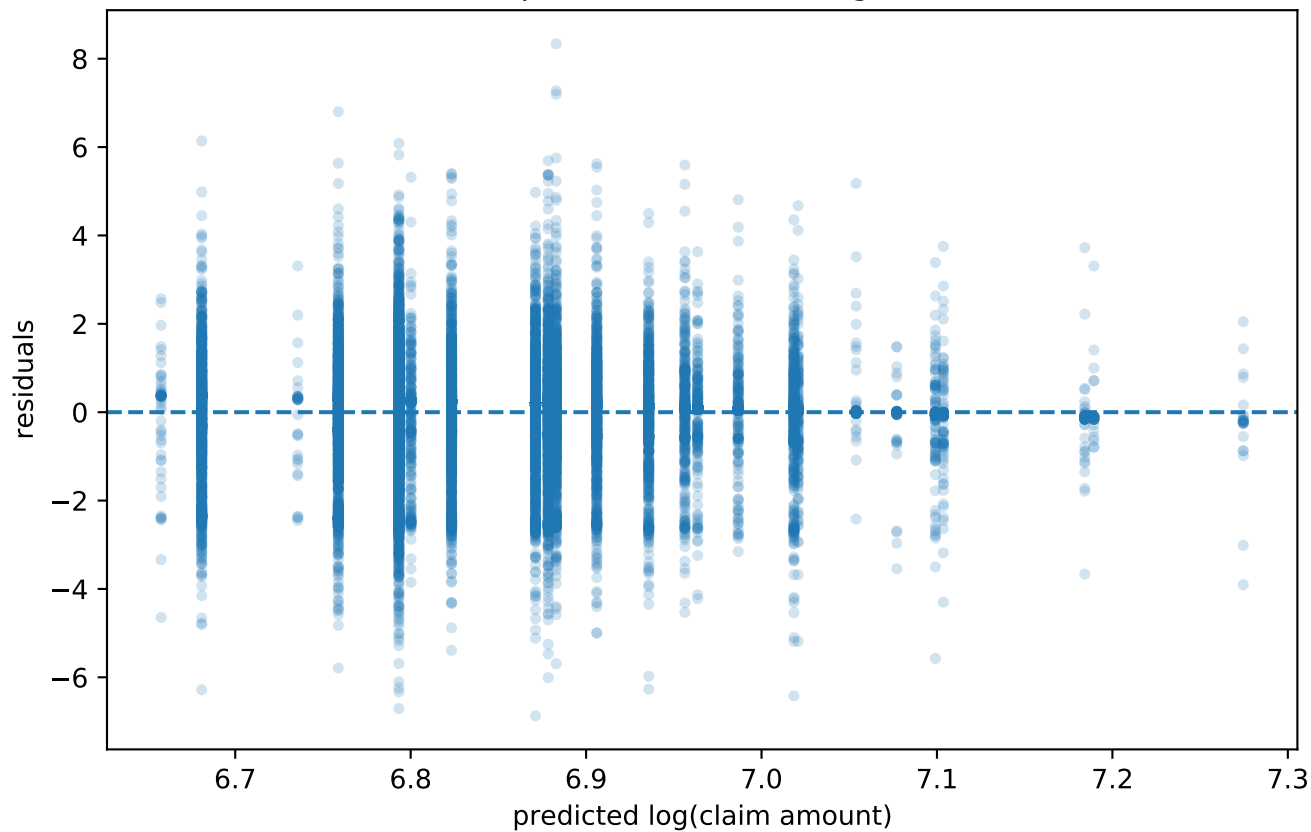
deviance residuals of gamma-GLM



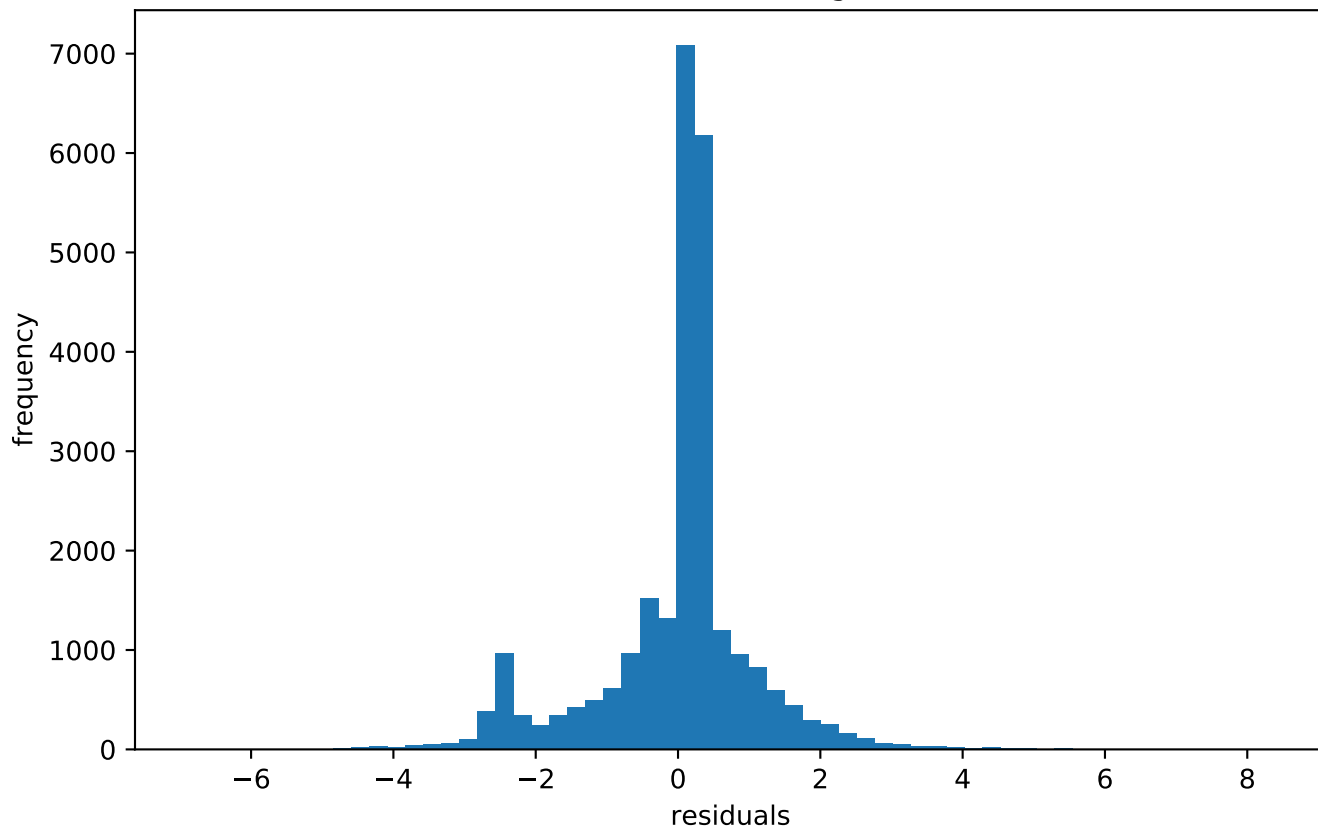
The scatter plot displays predicted claim amounts on the x-axis, ranging from 1500 to 5500. A horizontal dashed line is drawn across the plot at approximately y=1000. The data points are heavily clustered around this line, particularly between 1500 and 2500. There are several outliers above the line, with the most prominent one at approximately (5400, 5500). The plot shows a high density of points, suggesting a large dataset.



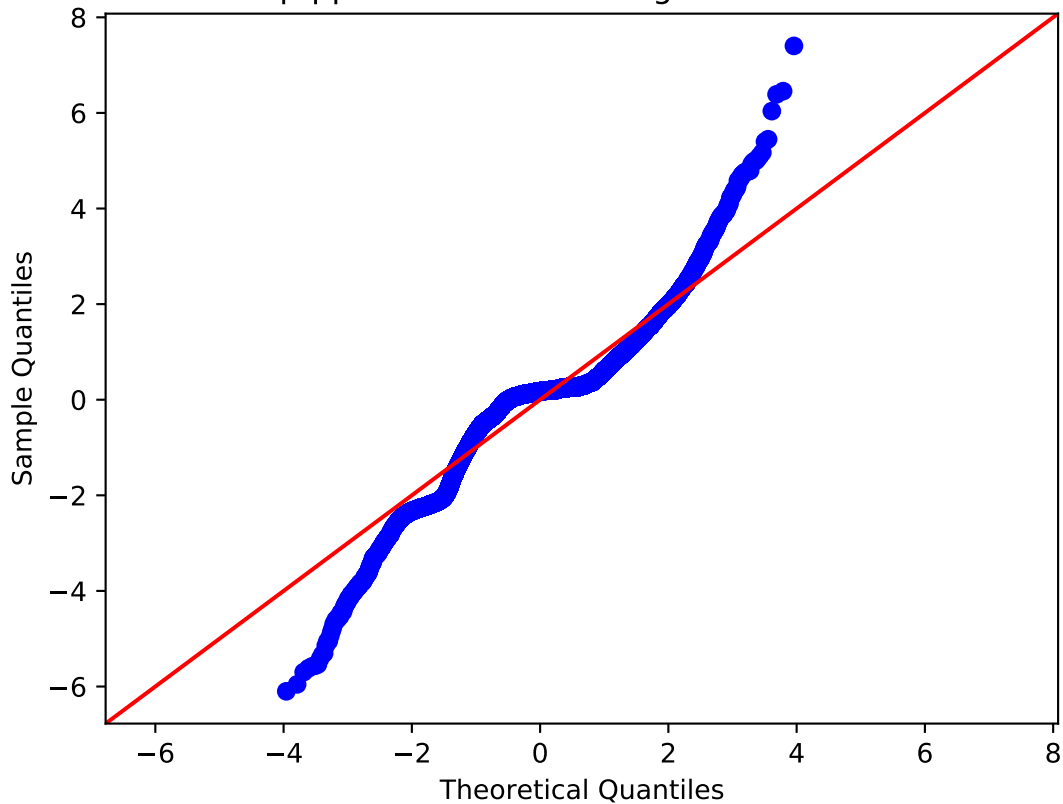
residuals vs. predicted values - lognormal-model



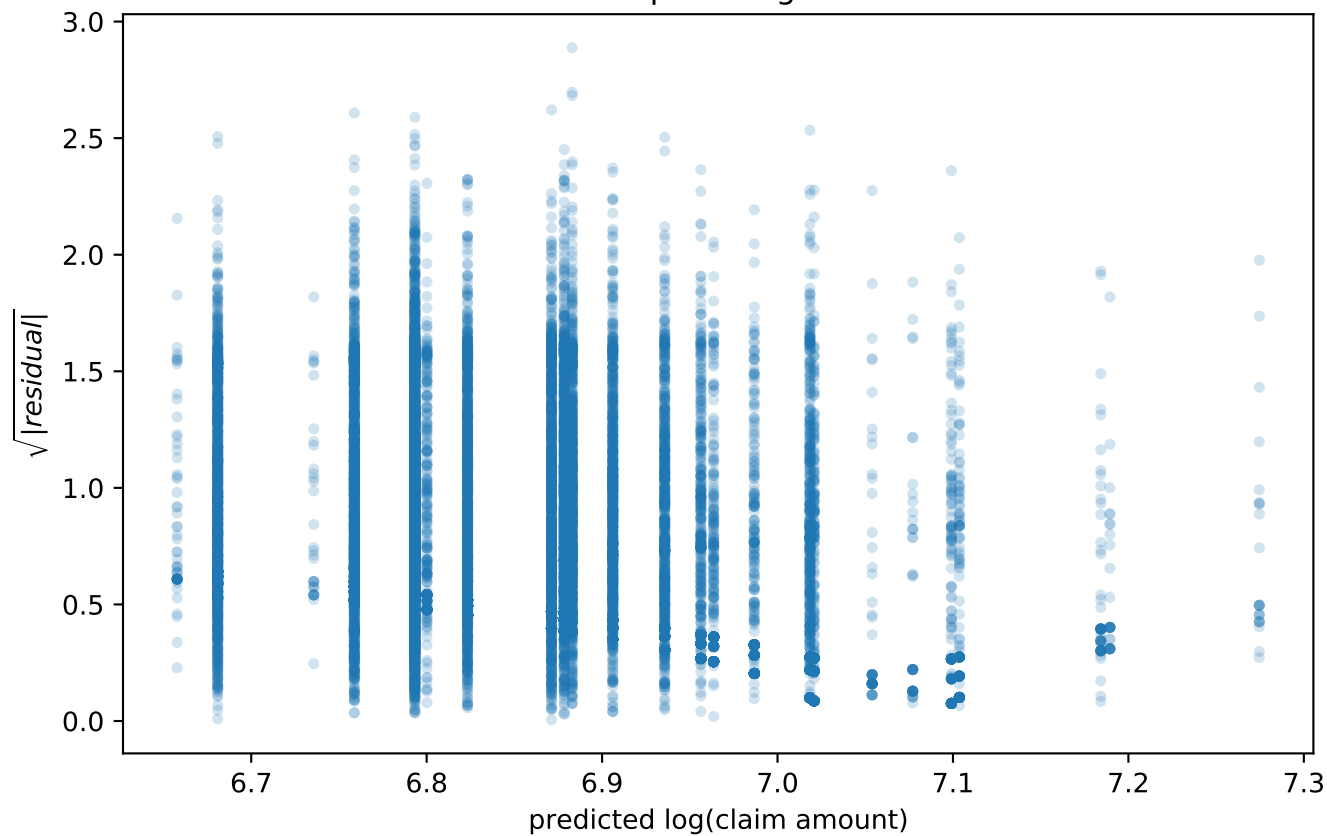
distribution of residuals - lognormal-model



q-q-plot of residuals - lognormal-model



scale-location-plot - lognormal-model



model comparison by age_group

	observed_mean	gamma_mean	lognormal_mean	count
18-25	5121.62	4826.9	1842.49	2403.0
26-40	2115.68	2139.58	1702.28	8179.0
41-60	1797.83	1805.61	1777.12	11893.0
61+	2246.47	2255.38	1923.5	3969.0

$$WMRAE = \sum w_g \left| \frac{y_g - y_o}{y_g} \right|$$

WMRAE (Age Groups) = 1.13%

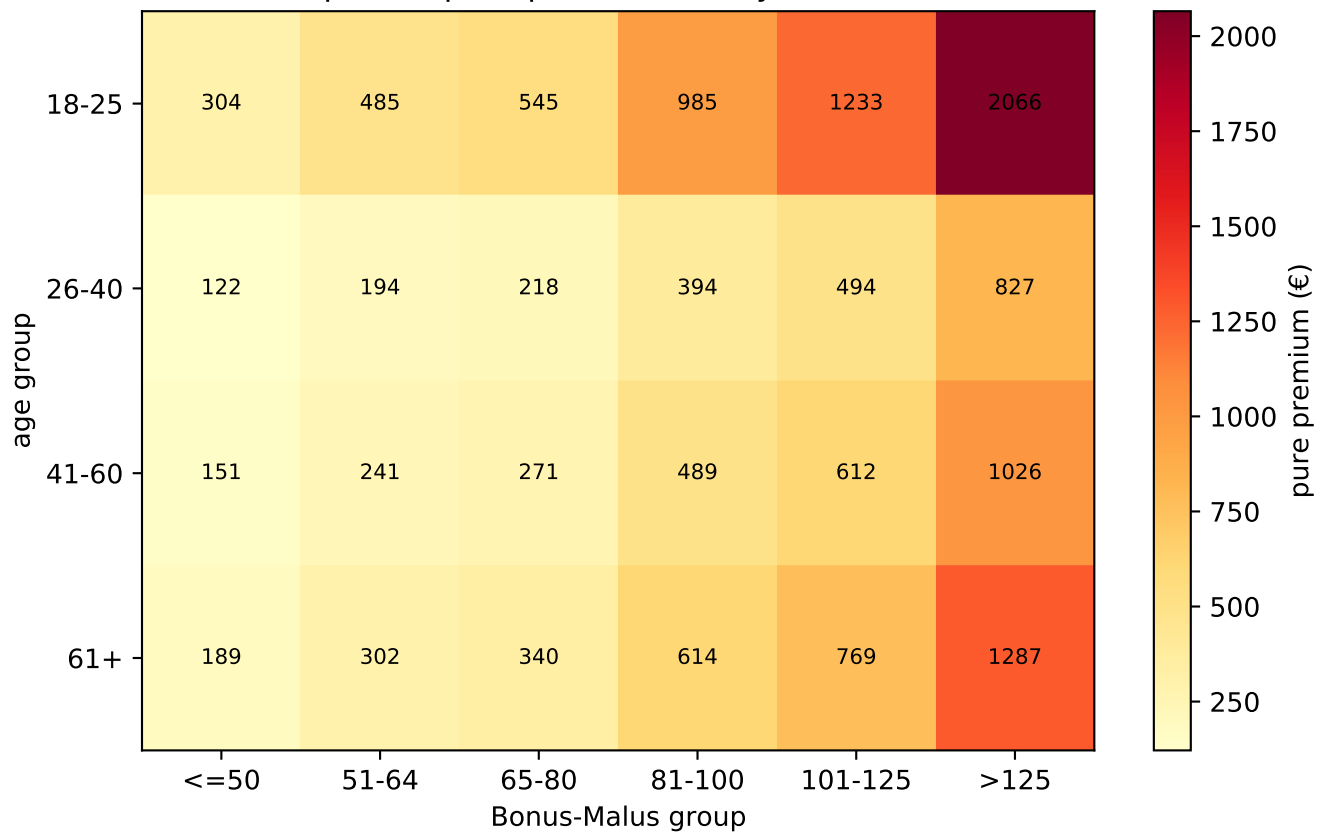
model comparison by bm_group

	observed_mean	gamma_mean	lognormal_mean	count
<=50	1922.22	1935.55	1687.83	11592.0
51-64	2022.05	2025.75	1757.43	5516.0
65-80	2039.67	2124.38	1829.31	3974.0
81-100	3849.88	3544.64	1928.76	4106.0
101-125	1844.08	2209.53	2094.51	1093.0
>125	3339.58	3078.19	2345.35	163.0

$$WMRAE = \sum w_g \left| \frac{y_g - y_o}{y_g} \right|$$

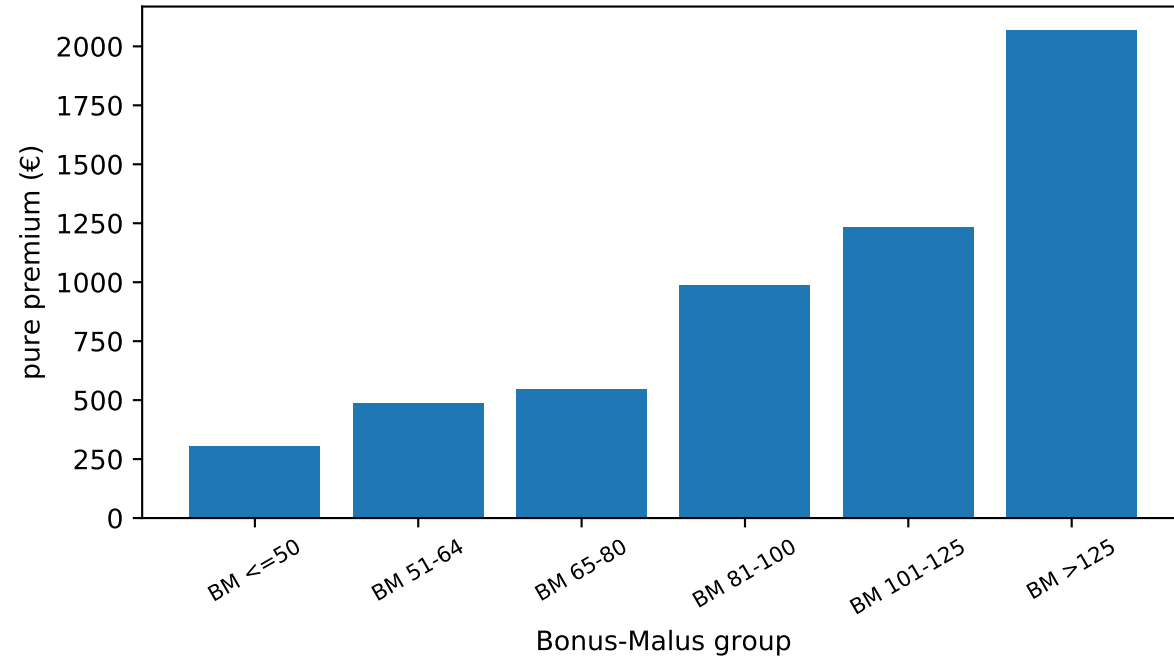
WMRAE (Bonus-Malus Groups) = 3.06%

expected pure premium (€) by tariff class

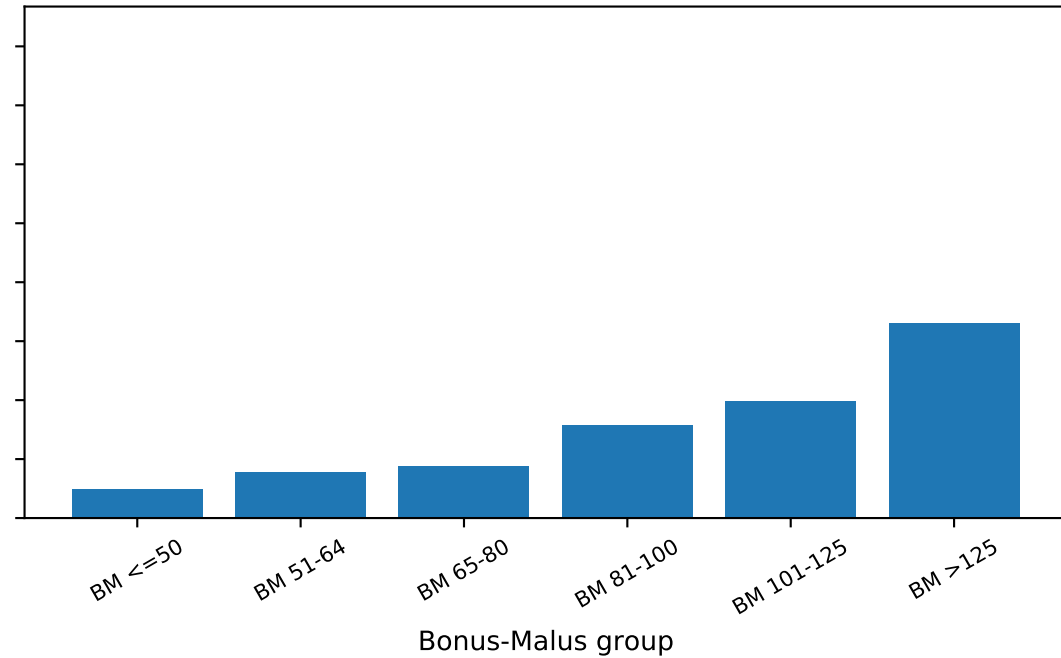


expected pure premium by age group and Bonus-Malus group

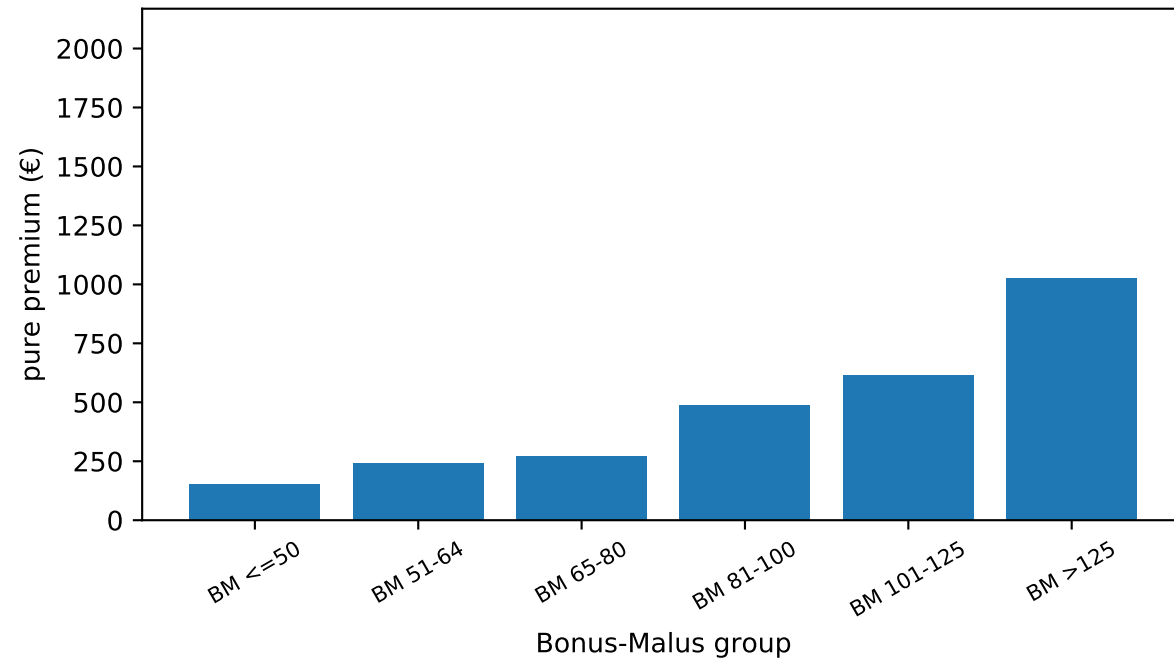
age 18-25



age 26-40



age 41-60



age 61+

